Paper 1

1 **B** How close the measured values of X are to its true value reflects the extent of the systematic error. The bigger $(X - x_0)$ is, the bigger the systematic error.

The amount of random error is reflected in how scattered the measured values are around a mean value. The wider the spread, the bigger the random error.

The graph in B shows that the mean value of X is far from the true value x_0 indicating a large systematic error, while the small spread of values about the mean value of X indicates a small random error.

2 **D** $\%\rho = 2(\%d) + \%V + \%I + \%l$